

PROJECT DESIGN PHASE-II

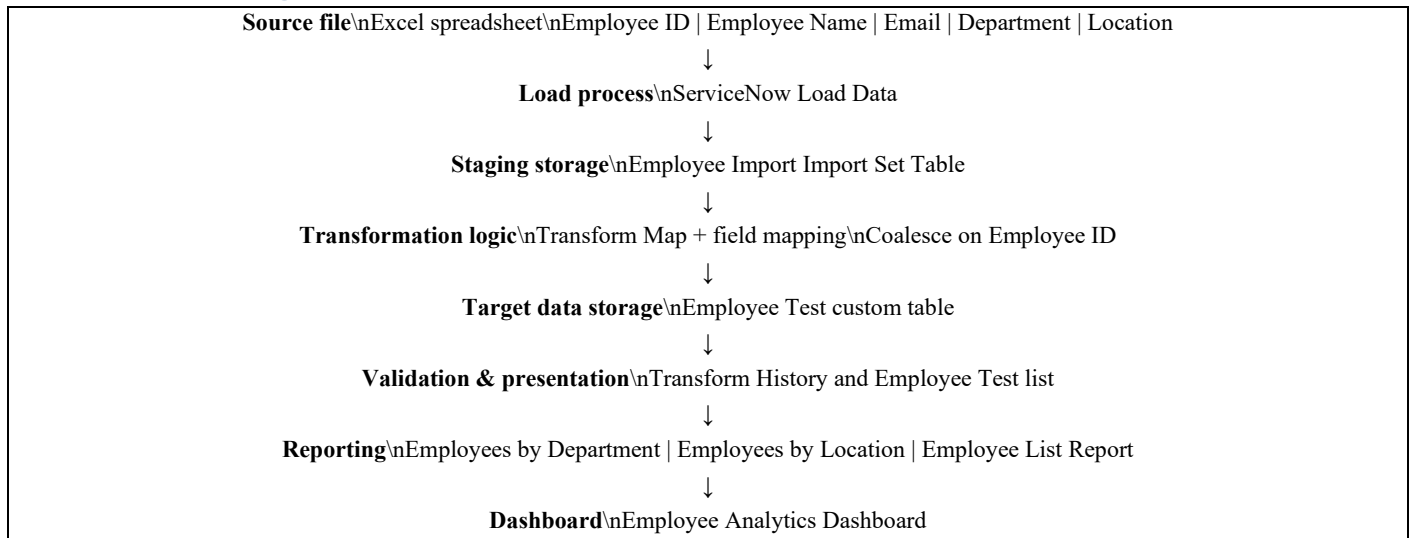
TECHNOLOGY STACK (ARCHITECTURE & STACK)

Team ID	SWTID-2026-1448
Project Name	Import Data Using Transform Maps (Spreadsheet)

Technical Architecture

The project uses ServiceNow to import employee records from an Excel spreadsheet, stage the rows in an Import Set table, transform and map the data into a custom Employee Test table, and present the resulting records through reports and a dashboard. The workflow is implemented within the ServiceNow platform; the Excel file is the input source.

Architecture Diagram



Data flow and technical architecture of the ServiceNow employee spreadsheet import project.

Infrastructure demarcation: ServiceNow-hosted platform. Input data is supplied as an Excel file and uploaded through Load Data. No separate third-party API, machine-learning model, or independently configured local/cloud server is specified in the project workflow.

Table-1: Components & Technologies

S.No.	Component	Description	Technology
1	User Interface	ServiceNow forms, application navigator, lists, reports, and dashboard used by the project user.	ServiceNow UI
2	Application Logic-1	Load spreadsheet rows into a staging import table.	ServiceNow Load Data / Import Sets
3	Application Logic-2	Map source spreadsheet columns to Employee Test fields and execute transformation.	ServiceNow Transform Maps / Field Maps
4	Application Logic-3	Match employee records by Employee ID; update matching records and insert new IDs.	Transform Map Coalesce
S.No.	Component	Description	Technology

5	Database / Target Table	Stores Employee ID, Employee Name, Email, Department, and Location.	ServiceNow custom table: Employee Test (u_employee_test)
6	Import / Staging Storage	Temporarily holds imported spreadsheet rows before transformation.	ServiceNow Import Set table: Employee Import (u_employee_import)
7	File Storage	Source data is selected and uploaded as an Excel workbook.	Excel .xlsx file; upload through Load Data
8	External API-1	No external API is described in the supplied project workflow.	Not used / not specified
9	External API-2	No second external API is described in the supplied project workflow.	Not used / not specified
10	Machine Learning Model	No machine-learning model is included in this project.	Not applicable
11	Infrastructure (Server / Cloud)	Application, import processing, target table, reports, and dashboard operate within the ServiceNow platform.	ServiceNow platform; hosting configuration not specified

Table-2: Application Characteristics

S.No.	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source framework is identified in the provided workflow; the project uses platform-native configuration.	ServiceNow platform features
2	Security Implementations	Access depends on ServiceNow roles, table permissions, and instance configuration. Specific ACL, encryption, or firewall settings are not documented in the workflow.	ServiceNow access controls; detailed configuration not specified
3	Scalable Architecture	The import process can be repeated with later spreadsheet files by reusing the import table and Transform Map. Larger volumes or performance limits are not quantified.	Import Sets and Transform Maps
4	Availability	The project runs on the ServiceNow platform. Specific availability targets, redundancy, or load-balancer settings are not described.	ServiceNow platform
5	Performance	Bulk loading and transformation reduce manual row-by-row entry. No throughput, requests-per-second, cache, or CDN measurements are specified.	Load Data, Import Sets, Transform Maps

References

Project workflow reference: Import Data using Transform Maps (Spreadsheet), supplied project document.